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Course: MCA412-2 - Research Methodology
LAB Exercise V –
Literature Review.

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Lab 5

Lab Exercise V:
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1. Literature Review

Pandi et al. [1] investigated the empirical relationship between in situ remote sensing reflectance (R_{rs}) and the cell density of mixotrophic *Noctiluca scintillans* in the northeastern Arabian Sea. Using hyperspectral radiometer profiles and surface bottle collections aboard FORV *Sagar Sampada*, the authors demonstrated that the reflectance band ratio R_{rs}(531)/R_{rs}(510) strongly correlates with cell abundance ($R^2 = 0.96$, $p < 10^{-3}$). Their findings confirmed that the optical reflectance peak at 531 nm paired with the 510 nm absorption trough isolates *Noctiluca* endosymbionts from background diatoms and suspended matter, highlighting the need for dedicated 510 nm band capabilities on future ocean-color satellites.

Shi and Wang [2] evaluated the seasonal and interannual dynamics of phytoplankton biomass across the Arabian Sea using VIIRS observations from 2012 to 2019 alongside historical SeaWiFS datasets. By integrating satellite chlorophyll-a with physical reanalysis fields from GODAS, the study quantified two distinct monsoon-driven mechanisms: summer blooms in the western basin are forced by coastal upwelling and upward Ekman pumping ($w \approx 1\ \mathrm{m/day}$ at 70 m depth), whereas winter blooms in the northern basin are triggered by convective overturning and mixed-layer deepening to ~80 m. Their comparative analysis revealed no significant long-term secular trend in Arabian Sea chlorophyll-a biomass over the two-decade record.

Kuttippurath et al. [3] analyzed two decades (1998–2019) of satellite chlorophyll-a and net primary productivity (NPP) in the northern Indian Ocean to determine the impact of atmospheric dust and warming trends. Using merged OC-CCI observations, MERRA-2 aerosol optical depth (AOD), and HYSPLIT air-mass backward trajectories, the authors detected a basin-wide NPP decline of $-0.048\ \mathrm{mg\ m^{-3}\ day^{-1}\ yr^{-1}}$ driven primarily by a $+0.02^\circ\mathrm{C\ yr^{-1}}$ sea surface warming trend. Furthermore, a 10% to 20% shift in air-mass transport from surrounding Middle Eastern deserts modulated trace mineral deposition, showing that atmospheric inputs can stimulate episodic blooms when upwelling is weak.

Li et al. [4] assessed the radiometric accuracy of VIIRS Level-2 remote sensing reflectance products across a 10-year period (2012–2021) using the global OC-CCI in situ benchmark. The study revealed that conventional near-infrared atmospheric correction algorithms experience nearly a 50% accuracy degradation when the solar zenith angle exceeds 70°, inducing large negative biases in the blue wavebands. To resolve this geometric limitation, the authors developed a neural network atmospheric correction model (NN-V) that reduced relative percentage differences from -38.59% to -5.27% and expanded valid wintertime observation coverage in high-latitude and low-sun conditions by nearly twentyfold.

Cael et al. [5] examined a 20-year MODIS-Aqua time series (2002–2022) to evaluate whether climate-change-driven signals are detectable in multi-band remote sensing reflectance rather than scalar chlorophyll-a. By applying multivariate linear regression combined with the Cochrane-Orcutt procedure to filter temporal autocorrelation, the authors identified statistically significant long-term trends across 40% of the global surface ocean, demonstrating that low-latitude waters are becoming greener. Their findings showed that multivariate R_{rs} captures ecological shifts and community restructuring significantly faster than univariate chlorophyll, which remains constrained by high natural interannual variability.

Dutkiewicz et al. [6] simulated twenty-first-century changes in ocean optical properties and community structure under a business-as-usual emissions scenario using a coupled ecosystem-radiative transfer model. Their analysis demonstrated that remotely sensed reflectance in the blue-green spectrum (specifically 487–512 nm) has lower interannual variability and exhibits a much shorter time of emergence (ToE) than scalar chlorophyll-a. The model indicated that shifts in phytoplankton functional groups, cellular pigment packaging, and colored dissolved organic matter (CDOM) bleaching alter spectral reflectance ratios, warning that fixed empirical chlorophyll algorithms will become inaccurate in a changing ocean.

Sathyendranath et al. [7] detailed the comprehensive development and validation of the multi-mission Ocean Colour Climate Change Initiative (OC-CCI) product suite spanning from 1997 to 2018. The methodology employed Quasi-Analytical Algorithm (QAA) band-shifting to map radiances from SeaWiFS, MODIS, MERIS, and VIIRS to unified reference wavebands, corrected inter-sensor spatial and seasonal biases, and implemented fuzzy-logic optical water classification. The resulting climate data record minimized multi-satellite discontinuities, significantly enhanced spatial coverage using the POLYMER atmospheric correction, and generated per-pixel uncertainty estimates based on extensive in situ matchups.

Hammond et al. [8] developed a hierarchical Bayesian spatio-temporal model to quantify regional surface chlorophyll-a trends and uncertainties using the 20-year OC-CCI dataset. By incorporating CMIP5 biogeochemical model projections as Bayesian priors to constrain trend estimates, the authors evaluated 23 discrete ocean provinces while filtering AR(1) temporal autocorrelation and spatial distance decay. While their weighted global average trend showed no significant net change ($0.08 \pm 0.35\%\text{ yr}^{-1}$), 18 of the 23 individual regions exhibited significant trends ($\pm 1.8\%\text{ yr}^{-1}$), underscoring the critical need to analyze marine ecosystem shifts at regional scales rather than global aggregations.

Bisson et al. [9] validated spaceborne particulate backscattering (b_{bp}) retrievals from MODIS, VIIRS, and OLCI against 2,486 coincident in situ observations acquired from autonomous Biogeochemical Argo (BGC-Argo) profiling floats at 700 nm. Evaluating the GIOP, QAA, and GSM inversion algorithms, the authors found that satellite-to-float b_{bp} median ratios ranged from 0.77 to 1.60, with performance degrading noticeably in regions with high spatial reflectance variance. The study proved that assuming a fixed backscattering spectral slope ($\gamma = 1$) in the GSM model causes up to 50% overestimation in low-biomass waters, emphasizing that variable spectral slope formulations are required for accurate carbon stock estimation.

Castant et al. [10] reconstructed a 4D, gap-free global dataset (latitude, longitude, depth, time) of Photosynthetically Active Radiation (PAR) and its diffuse attenuation coefficient ($K_d\text{PAR}$) from 1958 to 2022 at a 0.25° resolution. The authors utilized GlobColour satellite products combined with the Environmental String Model (ESM) and DINEOF interpolation driven by ERA5 atmospheric variables to reconstruct light penetration through the upper 250 m. Their validation against SEABASS in situ radiometric stations and biogeochemical hindcast models demonstrated that coupling machine learning with physical atmospheric reanalysis accurately reconstructs the depth of the euphotic zone (Z_{eu}) across decadal ocean cycles.

2. Summary and Findings

After reviewing these ten studies, it is clear that while satellite remote sensing and multi-mission ocean color datasets have come a long way, several key practical problems still remain unresolved. The biggest issue is that standard chlorophyll algorithms treat all green patches the same, frequently misclassifying complex mixotrophic dinoflagellate blooms like Noctiluca scintillans as ordinary diatom populations because they only check broad blue-to-green reflectance ratios. Most monitoring workflows also view satellite images as flat, isolated snapshots, completely ignoring essential subsurface physical processes like mixed-layer deepening, upwelling velocity, and atmospheric dust inputs that actually trigger these blooms in the first place. On top of that, satellite data often gets heavily degraded or masked during winter and low-sun observation windows, while traditional time-series methods that rely only on single-channel chlorophyll fail to catch real ecological shifts due to high natural noise and autocorrelation. Recent advancements show that we can overcome these barriers by designing targeted band ratios near 510 nm and 531 nm, using neural networks to clean up atmospheric errors, and switching from single chlorophyll numbers to multivariate reflectance models. Bringing these physical, atmospheric, and spectral layers together into one data pipeline offers a far more sensitive and reliable way to track harmful blooms and understand how our coastal ecosystems are genuinely shifting over time.

6. References

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